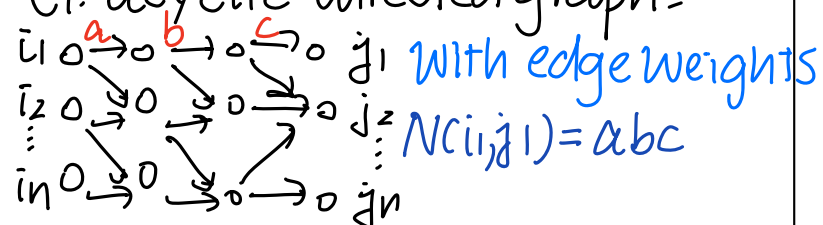


Appendix:

• Lindström-Gessel-Viennot lem)

G : acyclic directed graph.



$NC(i_1, \dots, i_p; j_1, \dots, j_p)$ = Weighted sum of **non-intersecting** paths from $i_1, \dots, i_p$ to $j_1, \dots, j_p$. Then:

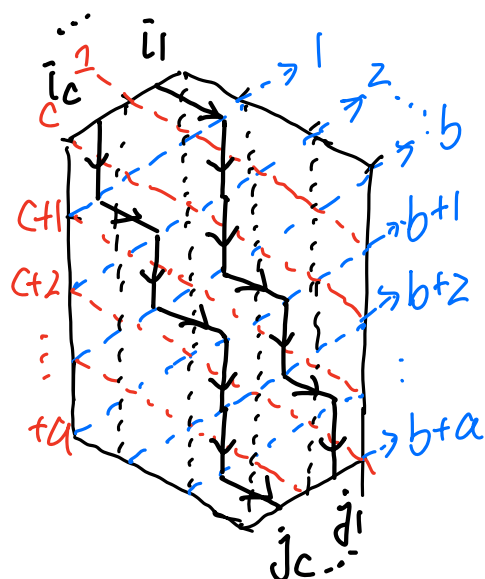
$$NC(i_1, \dots, i_n; j_1, \dots, j_n) = \det (NC(i_p, j_q))_{p,q=1}^n$$

• lemma (V)

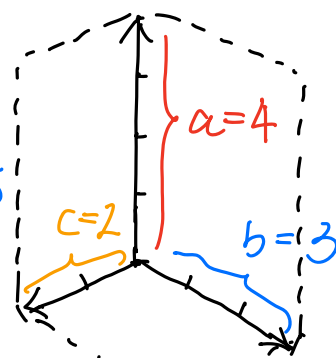
$$S_\lambda(1, q, \dots, q^{m-1}) = q^{\sum_{i=1}^m (i-1)\lambda_i} \prod_{1 \leq i < j \leq m} \frac{1 - q^{\lambda_i - \lambda_j + j - i}}{1 - q^{j-i}}$$

Goal: Count # of ways to stack boxes in $(a \times b \times c)$ corner.

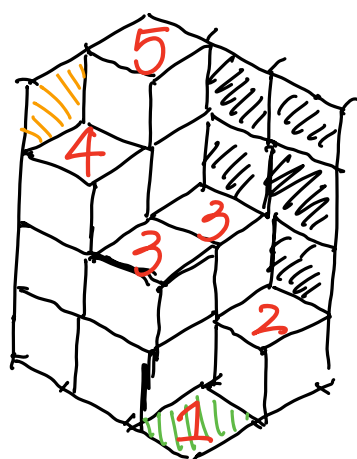
Method: Turn into NILP model



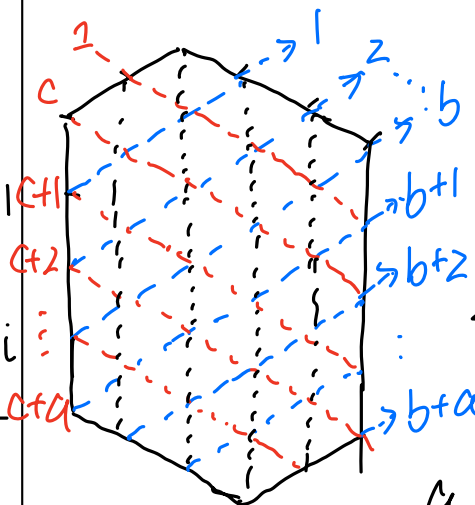
I.] Plane Partitions



$(a \times b \times c)$ Corner



Stack of boxes
rhombus tiles



$(a \times b \times c)$
hexagonal
region in
a triangular grid

A path $i_p \rightarrow j_q$ must have
 $(1 \leq p, q \leq c)$

$(a + q - p)$ many S-edges
 $(b + p - q)$ many SE-edges.

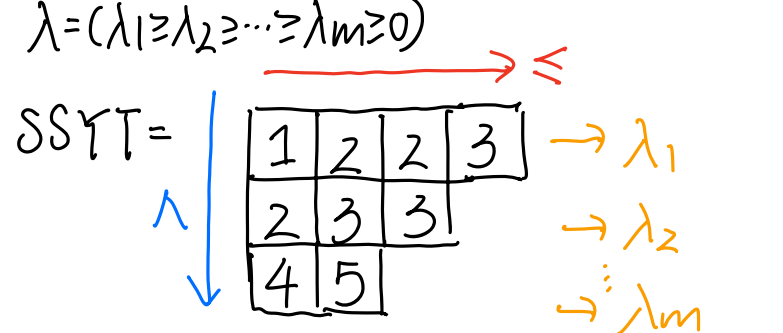
So: $NC(i_p; j_q) = \begin{pmatrix} a+b \\ a+q-p \end{pmatrix}$

↓ LGV

$$Na,b,c = \det \left[\begin{pmatrix} a+b \\ a+q-p \end{pmatrix} \right]_{p,q=1}^c$$

II. Background on Schur Functions

Semi-std Young Tableaux



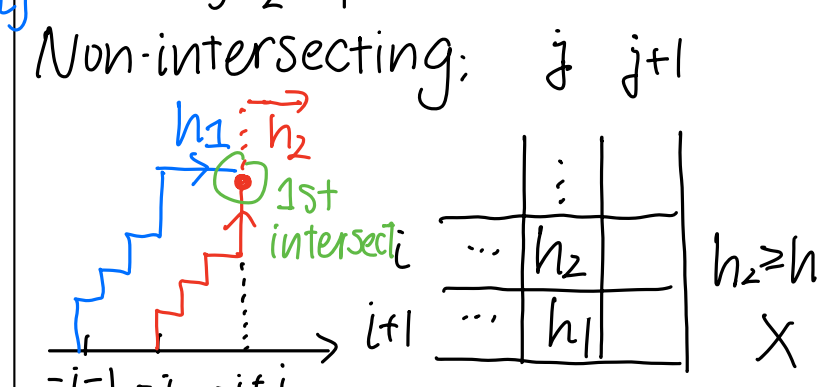
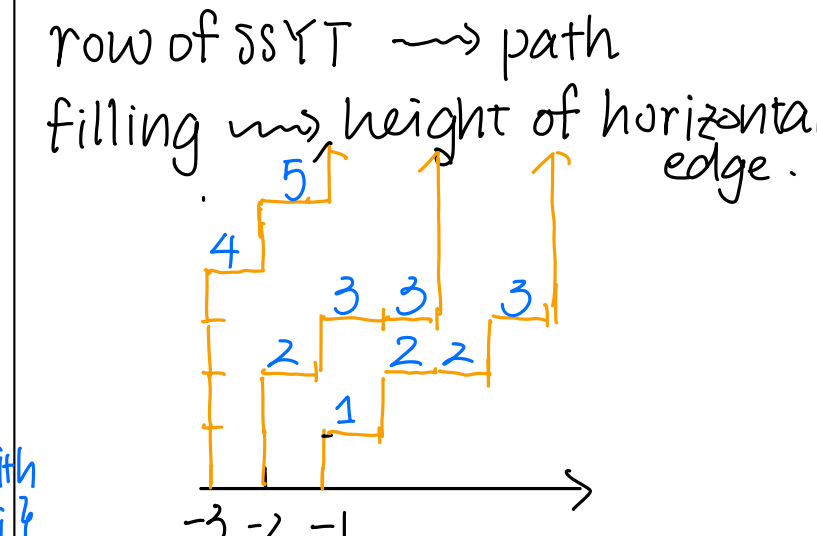
Schur Function.

$S_\lambda(x_1, \dots, x_n) = \sum_{\substack{\text{SSYT } p \\ \text{shape } \lambda \\ \text{fillings } \leq n}} \prod_{i=1}^n x_i^{\# \{ \text{boxes with filling } i \}}$

Example: $x_1^2 x_2^3 x_3^3 x_4 x_5$

We now present 2 NILP models

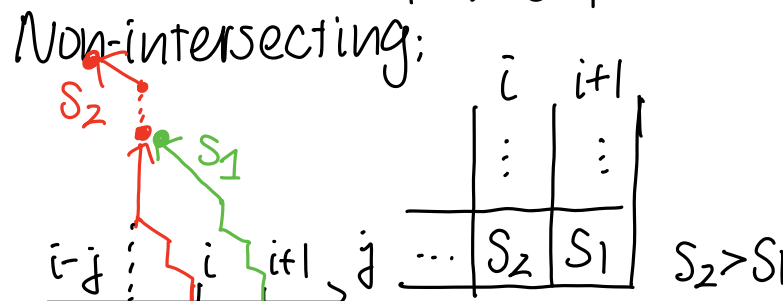
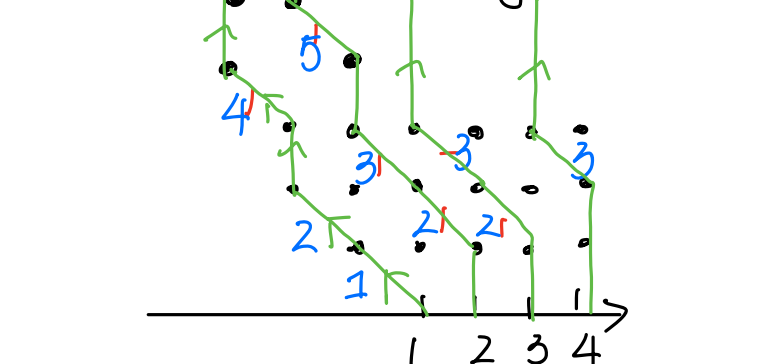
Model I, "Staircase Model".



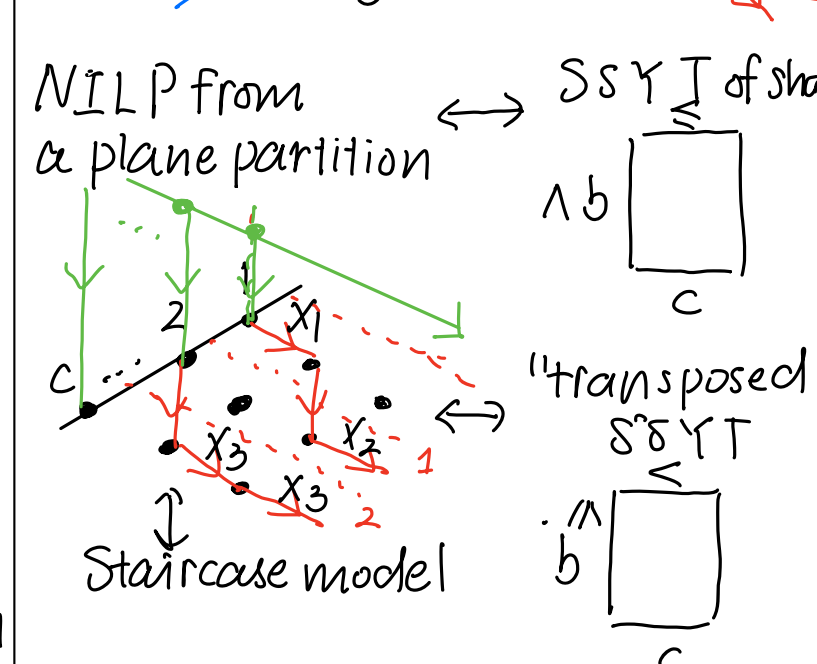
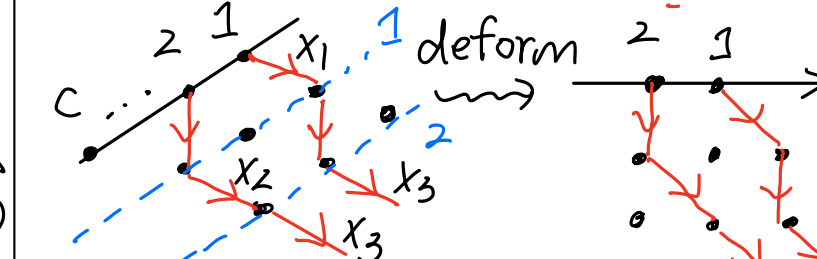
Model 2. "Slider Model".

col of SSYT $\rightsquigarrow$ path

filling of box $\rightsquigarrow$ step index of (or height of endpt)



III. Back to Enumeration of PP.



Prop. Put weight X_i on SE-edge with step index i :

$$N_{a,b,c}(X_1, \dots, X_{a+b}) = \sum_b \square_c (X_1, X_2, \dots, X_{a+b})$$

Thm (MacMahon Formula)

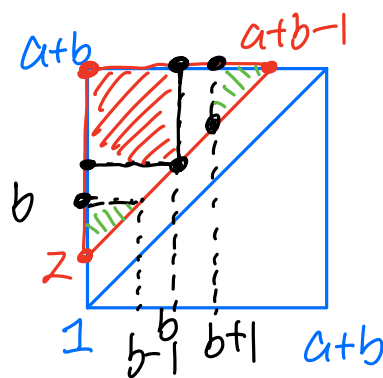
Let $X_i = q^{a+b-i}$. then

$$N_{a,b,c} = q^{(\text{const})} \prod_{i=1}^a \prod_{j=1}^b \prod_{k=1}^c \frac{1 - q^{i+j+k-1}}{1 - q^{i+j+k-2}}$$

Pf Sketch: $\text{len}(V) \Rightarrow$
 $N_{a,b,c}(q) = q^{\text{const}} \prod_{1 \leq i < j \leq a+b} \frac{1 - q^{\lambda_i - \lambda_j + j - i}}{1 - q^{j-i}}$

$$\lambda = (\underbrace{c, c, \dots, c}_b, \underbrace{0, 0, \dots, 0}_a)$$

$$\lambda_i - \lambda_j = \begin{cases} 0, & \text{if } j \leq b, \text{ or } i \geq b+1 \\ c, & \text{else.} \end{cases}$$

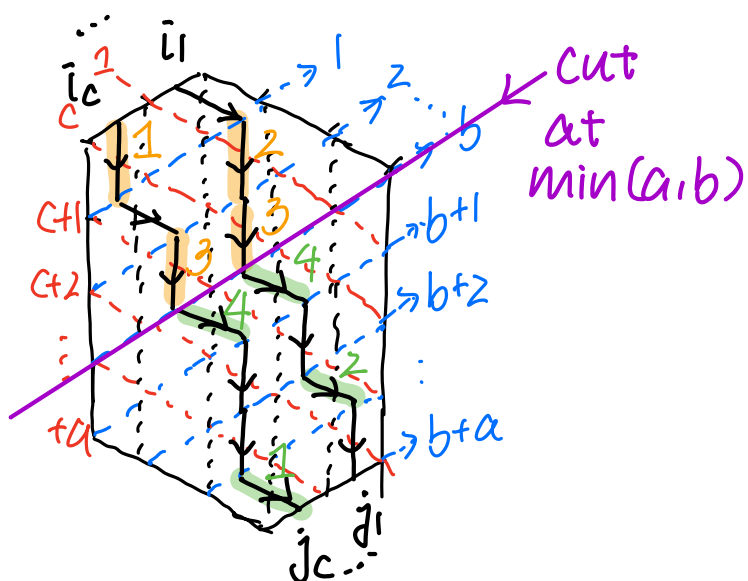


$$\begin{aligned} & \leadsto \prod_{i=1}^b \prod_{j=b+1}^{b+a} \frac{1 - q^{c+j-i}}{1 - q^{j-i}} \\ & = \prod_{i=1}^b \prod_{j=b+1}^{b+a} \left(\prod_{k=1}^c \frac{1 - q^{k+j-i}}{1 - q^{k+j-i-1}} \right) \end{aligned}$$

$q=1$ gives classical formula

$$N_{a,b,c} = \prod_{i=1}^a \prod_{j=1}^b \prod_{k=1}^c \frac{i+j+k-1}{i+j+k-2}$$

Another Schur Formula.



S-edges above interface = # SE edges below interface

$$\left(\begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & 3 \\ \hline \end{array} \right), \left(\begin{array}{|c|c|} \hline 1 & 2 \\ \hline 4 & 4 \\ \hline \end{array} \right)$$

$$\leq a \wedge b \left(\begin{array}{|c|c|} \hline \lambda \\ \hline \end{array} \right) \leq a \wedge b \left(\begin{array}{|c|c|} \hline \lambda \\ \hline \end{array} \right)$$

contents $\leq b$ contents $\leq a$

Prop.

$$N_{a,b,c} = \sum_{\lambda: \lambda_1 \leq c} S_{\lambda}(\overbrace{1,1,\dots,1}^a) S_{\lambda}(\overbrace{1,1,\dots,1}^b)$$

Running Ex Produces 2 Tableaux

